UGPAYMENTS – EXTERNAL MPI

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REVISION HISTORY

Date	Revision	Description	Author
7/15/2025	1	First Revision	Unicorn
8/13/2025	1.2	Added Authorization endpoint	Unicorn

OVERVIEW

WHAT IS 3-D SECURE 2.0

As the share of e-commerce and especially mobile transactions is growing, the industry has been suffering from high chargeback ratios and low approval ratios compared to card present transactions. With the introduction of 3D-Secure 2.0, this gap should be minimized.

The SCA mandate requires 3D-Secure to be applied to all electronic payments – including proximity, remote and mobile payments – within the European Economic Area (EEA). The strong customer authentication mandate is complemented by some limited exemptions that aim to support a frictionless customer experience when transaction risk is low.

3D-Secure 2.0 supports the transfer of rich data for transactions enabling possible risk-based decisions regarding whether to authenticate or not. This means that based on certain criteria, the end user may or may not be asked to perform a 'challenge' to
ensure their authenticity. The consumer experience has also been simplified and enhanced thanks to the elimination of the preliminary enrolment process and by not requiring the end user to remember as many passwords.

During the 3D-Secure 2.0 authentication, the issuer performs a risk-based analysis to determine whether a transaction should be challenged. A challenge means that the user is presented with an authentication screen of the issuer that follows the Strong Customer Authentication rules. Issuer risk analysis is performed based on the data collected during the transaction, such as device information, transaction information, time zone and various other parameters. If an issuer determines that the transaction is low risk, a frictionless indicator is returned in the authentication response. Frictionless flow means that the transaction is considered 3D authenticated and a liability shift applies, without the need of the user challenge screen.

It is highly recommended that merchants send as many optional parameters as possible during the authentication, as it increases the chance of frictionless flow and better user conversion.

This endpoint allows you to use an external MPI, and send the successful authentication 3D values to our endpoint for processing mandated 3D transactions.

REQUIRED INFORMATION

Before you will be able to submit transactions to UnicornPayment, you will need an UnicornPayment merchant account for your website. Once you have a merchant account established, UnicornPayment will supply you with:

1. Merchant Id
2. Site Id
3. OAuth Bearer Token

These IDs uniquely identify your websites, customers, and payments.

SOFTWARE REQUIREMENTS

To implement the JSON interface for standard card processing, the following requirements must be met:

- Working knowledge of JSON
- SSL server supporting 128-bit (or stronger) encryption

EXTERNAL MPI TRANSACTIONS

OPERATION END-POINT

The URL to submit verify enrollment transactions is as follows:
https://api.ugpayments.ch/merchants/<merchantid>/externalmpisaletransactions

For authorizations, the URL will be:
https://api.ugpayments.ch/merchants/<merchantid>/externalmpiauthtransactions

VERIFY ENROLLMENT TRANSACTION REQUEST

The following table lists the elements for verify enrollment transactions:

Parameter Name	Description	Example
siteId	Site ID assigned by Unicorn Integer, mandatory	12
amount	Amount of the transaction Decimal, mandatory	123.45
currency	Transaction currency, in ISO 4217 format. 3 character, mandatory	USD
firstName	Cardholder's first name 50 character max, mandatory	John
lastName	Cardholder's last name 50 character max, mandatory	Doe
phone	Cardholder's phone 20 character max, mandatory	555-555-5555
addressLine1	Billing address 100 character max, mandatory	123 Fake St.
addressLine2	Billing address 100 character max, optional	Suite # 789
city	Billing address city	Hollywood

	50 character max, mandatory	
state	Billing state 50 character max, mandatory	California
countryId	Billing country 2 character, mandatory	US
postalCode	Billing postal code 20 character max, mandatory	90046
email	Cardholder's email address 100 character max, mandatory	<identifier>@<domain>.<extension>
cardNumber	Credit card number 50 character max, mandatory	4024007118676898
nameOnCard	Cardholder's name printed on card 40 character max, mandatory	John J. Doe
expirationMonth	2 character, mandatory	09
expirationYear	4 character, mandatory	2018
cVVCode	4 character max, mandatory	469
iPAddress	Cardholder's IP Address 20 character max, mandatory	127.0.0.1
trackingId	Merchant Provided number, example reference number or merchant's transactionID 100 character max, optional	123456
isInitialForRecurring	Indicates if the transaction is the initial recurring transaction, if recurring, send true, mandatory	true or false *PLEASE NOTE, IF SET TO TRUE THEN 3D WILL ALLOWS RESPOND WITH CHALLENGE REQUEST. IF SET TO FALSE, YOU MAY GET A FRICTIONLESS RESPONSE BACK.

ConfirmURL	Merchant's URL that can receive a postback/callback	https://www.merchanturl.com/postback.php
External MPI Parameters		
eci	Eci value 3 character max, mandatory	05
acsVerificationId	Same authentication value that is returned for a successful authentication 30 character max, mandatory	MTIzNDU2Nzg5MDEyMzQ1Njc4OTA=
dsTransId	DS Transaction ID 40 character max, mandatory	522ff8f1-04b8-4fac-95bb-ce60e0b322ad
threeDServerTransId	40 character max, optional, must send if available	522ff8f1-04b8-4fac-95bb-ce60e0b322ad
acsTransId	ACS Transaction ID, 40 character max, optional, must send if available	fea7c9eb-3016-497e-9114-8f2d82faf44f
threeDVersion	Version number of 3D 5 characters, mandatory	2.2.0
threeDAuthenticationStatus	Status of Authentication 1 character max, optional, must send if availabe	Y
ThreeDEnrollmentStatus	it can be assumed EnrollmentStatus as "Y" if the acsVerificationId, eci, dsTransId are available or the authentication is successful 1 character max, optional, must send if available	Y
MessageVersion	This is a standard 3Ds field set by EVMCo for all 3Ds protocol, so as other fields it	2.0

	may or may not available from the MPI5 character max, optional, must send if available	
IsExternalMPI	Required for all External MPI transactions, will always be True. Boolean	True

JSON Format Example:

{"siteId":"11",
"amount":123.45,
"currency":"USD",
"firstName":"John",
"lastName":"Doe",
"phone":"555-555-5555",
"addressLine1":"123 Fake St.",
"addressLine2":"",
"city":"Hollywood",
"state":"CA",
"countryId":"US",
"postalCode":"12345",
"email":"",
"cardNumber":4024007118676898,
"nameOnCard":"John J Doe",
"expirationMonth":11,
"expirationYear":2019,
"cVVCode":"402",
"iPAddress":"127.0.0.0",
"trackingID" : "123",
"isInitialForRecurring":false,
"ConfirmURL": https://www.merchanturl.com/post.php,
"eci":"05
"AcsVerificationId":"AJkCAhc5BgAAAADIhAGVdQAAAAA=",
"DsTransId":"b60a63f7-5760-4b6f-9d16-18684e6d23dc",
"threeDServerTransId":"",
"AcsTransId":"af60ce6e-e596-4e30-b674-cbb34d28fb89",
"ThreeDVersion":"2.2.0",
"ThreeDAuthenticationStatus":"Y",
"ThreeDEnrollmentStatus":Y,
"MessageVersion":"2.0",
"IsExternalMPI":True,
}

EXTERNAL MPI TRANSACTION RESPONSE

The result is in JSON format with the following elements:

Parameter Name	Description	Example
state	The type of transaction	Sale
status	The transaction status	Successful
message	The bank message	Transaction Suceeded
reasonCode	Any additional message with the transacton	00 - Approved

VOID TRANSACTIONS

URL: https://api.ugpayments.ch/merchants/[MerchantId]/voidtransactions
Only authorize transaction can be voided.

JSON REQUEST PARAMETERS

Parameter Name	**Description**	**Example**
authorizeTransactionid	The transaction id returned by the authorization. String, mandatory	123456

JSON RESPONSE PARAMETERS

Parameter Name	**Description**	**Example**
State	String, state of transaction, example Sale, Auth, Refund	Sale
Status	String, Status of transaction	Approved
message	String, message regarding the transaction	The operation was successfully processed.
trackingId	String, passed from request	YourID
id	String, transactionID from UnicornPayment	transactionID
reasonCode	String, Response From bank ONLY if transaction reached bank.	00 - Approved

HTTP REQUEST EXAMPLE

```
{
        "authorizeTransactionId": "123456"
}
```

HTTP RESPONSE EXAMPLE

```
{
      "id": "123457",
      "message": "Success",
      "state": "Void",
      "status": "Successful"
      "reasoncode": "00-approved"

}
```

3-D SALE REBILL

If you wish to rebill a previously authenticated transaction, you can use the /recurringtransactions endpoint.
You must use the Sale Transaction ID returned from the /ThreeDSaleTransactions endpoint.

If you wish to rebill any transaction, you must send 'IsInitialForRecurring' = 'true' on every endpoint on the 3D flow:

/verifytransactions
/threedsaletransactions

3D SALE REBILL REQUEST

The following table lists the elements for rebilling 3D transactions:

Parameter Name	Description	Example
TransactionID	The identifier of the sale transaction	123456
Amount	The amount of the transaction you wish to charge again	2.45
Currency	The currency of the transaction. This must match the original sale transaction currency	USD

3D SALE REBILL RESPONSE

The result is in JSON format with the following elements:

Parameter Name	Description	Example
Id	The id of the rebilled transaction	171888
Message	Message from bank response	APPROVED
State	The type of transaction. Please refer to Appendix A for detail.	Recurring
Status	The transaction status. Please refer to Appendix B for detail.	Successful

Example Response JSON:


```
{
  "state": "Recurring",
  "status": "Successful",
  "message": "Approved",
  "id": “171888"
}
```


CONFIRM PAGE

When a customer has successfully completed a transaction, UnicornPayment will send an HTTP POST with the transaction details back to a designated page on your site. This page should validate and store the transaction information in your database. The confirm page provides communication between UnicornPayment and your application and does not need to provide user functionality.
Information is posted to the confirm page as standard HTTP POST name-value pairs (NVP) separated by ampersands (&). An example confirm post would be:
Amount=17.99&MerchantReference=abc123&PayReferenceID=b9ab260b-d690-4507-8d56-8bd92c4c132a&TransactionID=4cfdefc3-6ad2-49de-a25b-5d0f41e8cd1a

The following fields are transmitted by UnicornPayment in a transaction to the ConfirmURL specified by you:

[TrackingId] => Reference Number for the Merchant’s Records (Limit 100 characters)

[MerchantReference] => Reference Number for the Merchant’s Records (Limit 100 characters)

[CurrencyID] => ISO Currency Code
[Key] => key supplied from portal from UnicornPayment

[Amount] => Amount of Transaction (Money/Numeric/Decimal)

[TransactionID] => Identity Number for the Transaction (Numeric)

[CardMask] => Characters used to mask the card number

[TransactionState] => State of Transaction (Appendix A)

[TransactionStatus] => State of Transaction (Appendix B)

[ShippingFirstName] => First Name for Shipping Address (Limit 50 characters)

[ShippingLastName] => Last Name for Shipping Address (Limit 50 characters)

[ShippingAddress1] => Address Line 1 for Shipping Address (Limit 100 characters)

[ShippingAddress2] => Address Line 2 for Shipping Address (Limit 100 characters)

[ShippingCity] => City for Shipping Address (Limit 50 characters)

[ShippingState] => State for Shipping Address (Limit 50 characters)

[ShippingCountry] => Country for Shipping Address

[ShippingPostalCode] => Postal Code for Shipping Address (Limit 20 characters)

[CustomerEmail] => mailto:mail@mail.com Email for the Customer (Limit 50 characters)

[CustomerFirstName] => First Name of Customer (Limit 50 characters)

[CustomerLastName] => Last name of Customer (Limit 50 characters)

[CustomerAddress1] => Address Line 1 for Customer Address (Limit 100 characters)

[CustomerAddress2] => Address Line 2 for Customer Address (Limit 100 characters)

[CustomerCity] => City for Customer Address (Limit 50 characters)

[CustomerState] => State for Customer Address (Limit 50 characters)

[CustomerCountry] => Country for Customer Address

[CustomerPostalCode] => Postal Code for Customer Address (Limit 20 characters)

[CustomerPhone] => Customer's Phone (Limit 20 characters)

[SiteID] => Site Id for the Merchant's Site (Numeric)

REFUND TRANSACTIONS

REFUND TRANSACTIONS

URL: https://api.ugpayments.ch/merchants/[MerchantId]/refundtransactions
Only sale and capture transactions can be refunded.

JSON REQUEST PARAMETERS

Parameter Name	Description	Example
referencetransactionid	The transaction id returned by the sale transaction or the capture transaction. String, mandatory	123456

amount	The amount to refund, not to exceed original transaction amount. Decimal/Number, mandatory	123.45

JSON RESPONSE PARAMETERS

Parameter Name	Description	Example
State	String, state of transaction, example Sale, Auth, Refund	Sale
Status	String, Status of transaction	Approved
message	String, message regarding the transaction	The operation was successfully processed.
trackingId	String, passed from request	YourID
id	String, transactionID from UnicornPayment	transactionID
reasonCode	String, Response From bank ONLY if transaction reached bank.	00 - Approved

HTTP REQUEST EXAMPLE

```
{
        "referenceTransactionId": "123456",
        "amount": 123.45
}
```

HTTP RESPONSE EXAMPLE

```
{
        "id": "123456",
        "message": "Success",
        "state": "Refund",
        "status": "Successful"
        "reasoncode": "00-approved"

}
```

APPENDIX A – TRANSACTION STATE TYPES

Sale
Authorize
Capture
Void
Refund

APPENDIX B – TRANSACTION STATUS TYPES

Successful
Error
Declined
Pending
Scrubbed
Fraud
Unconfirmed

APPENDIX C – VALIDATION ERRORS

ACS Verification Id is required when supplying ECI or XID.
Address Line1' can not contain a credit card number.
Address Line1' must be between 0 and 100 characters. You entered <char> characters.
Address Line1' must be between 1 and 100 characters.
Address Line2' can not contain a credit card number.
Address Line2' must be between 0 and 100 characters. You entered <char> characters.
Address Line2' must be between 1 and 100 characters.
Amount' must be greater than '0'.
Amount' must be greater than or equal to '0'.
Authorize Transaction Id' is not valid.
Bank Identifier' should not be empty.
Card Number' is not valid.
City' can not contain a credit card number.
City' must be between 0 and 50 characters. You entered <char> characters.
City' must be between 1 and 50 characters
Country Id' is not valid.

Country Id' must be not be empty.
'Currency' can not contain a credit card number.
Currency' is not valid.
'CVV code' is not valid.
ECI is required when supplying ACS Verification Id or XID.
Email' must be between 0 and 100 characters. You entered <char> characters.
Email' must be between 1 and 100 characters. You entered <char> characters.
Email' should not be empty.
Email' can not contain a credit card number.
Expiration date' is not valid.
First Name' can not contain a credit card number.
First Name' must be between 1 and 50 characters. You entered <char> characters.
First Name' should not be empty.
IP Address' can not contain a credit card number.
IP Address' must be between 0 and 20 characters. You entered <char> characters.
Last Name' can not contain a credit card number.
Last Name' must be between 1 and 50 characters. You entered <char> characters.
Last Name' should not be empty.
Merchant Referrer Url' must be between 0 and 100 characters. You entered <char> characters.
Name On Card' can not contain a credit card number.
Phone' must be between 0 and 20 characters. You entered <char> characters.
Phone' must be between 1 and 20 characters. You entered <char> characters.
Phone' should not be empty.
Postal Code' must be between 0 and 20 characters. You entered <char> characters.
Postal Code' must be between 1 and 20 characters in length.
Reason Id' is not valid.
Reference Transaction Id' is not valid.
Return Url' should not be empty.
'Sale Transaction Id' is not valid.
Shipping Address Line1' can not contain a credit card number.
Shipping Address Line1' must be between 0 and 100 characters. You entered <char> characters.

Shipping Address Line2' can not contain a credit card number.
Shipping Address Line2' must be between 0 and 100 characters. You entered <char> characters.
Shipping City' can not contain a credit card number.
Shipping City' must be between 0 and 50 characters. You entered <char> characters.
Shipping Country Id' can not contain a credit card number.
Shipping Country Id' is not valid.
Shipping First Name' can not contain a credit card number.
Shipping First Name' must be between 0 and 50 characters. You entered <char> characters.
Shipping Last Name' can not contain a credit card number.
Shipping Last Name' must be between 0 and 50 characters. You entered <char> characters.
Shipping Phone' must be between 0 and 20 characters. You entered <char> characters.
Shipping Postal Code' can not contain a credit card number.
Shipping Postal Code' must be between 0 and 20 characters. You entered <char> characters.
Shipping State' can not contain a credit card number.
Shipping State' must be between 0 and 50 characters. You entered <char> characters.
Site Id' can not contain a credit card number.
'Site Id' is not valid.
'State' can not contain a credit card number.
State' must be between 0 and 50 characters. You entered <char> characters.
State' must be between 1 and 50 characters.
Subscription plan id' is not valid.
Token Id' is not valid.
Tracking Id' must be between 0 and 100 characters. You entered <char> characters.
Validation error for input parameters.
Verify Transaction Id' is not valid.
XID is required when supplying ECI or ACS Verification Id.

APPENDIX D – SCRUB MESSAGES

Amount Limit
Approved Refferer Url Required
Card Bank ID Number IP Address
Card Decline Limiting

Card Holder Name
Card Limit
Country Black List
Credit Card Name and Customer Name Match
CVV2 Required
CyberSource
Email Limit
Email Required
Expired Credit Card
IP Address Bill
IP Black List
Name Credit Card Match
Name Declined Limit
Name Limit
Non US Customer
Phone Required
Price Limit
Price Minimum/Maximum
Refund Limit
Region Black List

APPENDIX E – ERROR MESSAGE

3D security is not enabled for this merchant.
Account associated with token is either deleted or disabled.
An address is required for this transaction.
An internal exception has occurred. Please refer to Error Id - <errorid>
Authorization headers not contained in the request.
Cannot do a full charge back for this transaction. Only partial charge back is allowed.
Cannot do a full refund for this transaction. Only partial refund is allowed.
Cannot do a full retrieval for this transaction. Only partial retrieval is allowed.

Capture amount is greater than the remaining authorization amount to be captured.
Capture is not enabled.
Cascade set up not valid for this transaction.
Client is not valid.
Credit card token used for transaction is no longer valid.
Critical error. Contact Unicorn.
Direct integration is not enabled.
Forced capture must have a bank reference as well.
Forced refund must have a bank reference as well.
Invalid Permissions.
Invalid principal identity.
Invalid transaction parameter(s).
Merchant does not exist or is no longer active.
Not a valid user.
Not able to capture. The transaction might be voided or already captured.
Not able to refund. The transaction might have CBK1 record or is already refunded.
Not authorized to access this resource.
Not authorized to execute this method.
Null Reason Code or specified Reason Code does not match the Card and Transaction Type thus cannot be authorized.
Partial charge back amount is greater than the remaining amount to be charged back.
Partial refund amount is greater than the remaining amount to be refunded.
Partial retrieval amount is greater than the remaining amount to be retrieved.
Rates not available for credit card and currency type.
Refund is not enabled.
Request is not in correct format.
Service connection failed.
Site does not belong to the merchant.
Site does not exist or is no longer active.
Subscription plan is not valid.
Test card is not enabled.

The authenticaiton token in no longed valid.
The credit card number is blacklisted.
The site is missing required bank parameter.
The specified transaction is not valid for Blacklisting.
The specified transaction is not valid for CBK1 Credit.
The specified transaction is not valid for Charge back.
The specified transaction is not valid for Dispute.
The specified transaction is not valid for Void.
The specified transaction is not valid for Whitelisting.
The volume limit for transactions have been exceeded.
This account is currently disabled, please contact risk@unicornpayment.com with any questions.
This function is not supported by the bank.
This transaction has already been captured.
This transaction is already been charged back.
This transaction is already been initiated for retrieval.
This transaction is already been refunded.
This transaction is already been voided.
Time period to capture this transaction has expired.
Transaction id is not valid.
Transaction is declined for failed 3D authentication. 3D authentication is required for this merchant.
Unable to locate successful transaction details.
User is disabled or is no longer active.
User not mapped to any account.
Virtual terminal is not enabled.